ROMAN URDU COMPILER

Compiler Construction Project Idea

Project Members	
Muhammad Faizan	23L-0567
Maahi Ibn Ahmar	23L-0990
Mahad Bin Atif	22L-4951

1. Project Idea

Roman Urdu Compiler is a small programming language designed for students who are comfortable with Roman Urdu but are still learning formal programming syntax. Instead of using only English-based keywords and symbols, the language uses simple Roman Urdu words for common programming operations. For example, the semicolon used to terminate a statement in C/C++ can be replaced by the keyword kardo.

2. Purpose of the Language

The main purpose of the language is to make introductory programming more natural and approachable for Urdu-speaking learners. It keeps the basic structure of a compiled programming language while allowing commands to be written using familiar Roman Urdu words. This project is mainly educational and demonstrates the complete compiler pipeline from source code to LLVM intermediate representation.

3. Key Features

• Roman Urdu keywords: programming actions use readable words such as agar (if), warna (else), jabtak (while), dikhao (print), and wapis (return).

• Statement terminator: kardo is used instead of the usual C/C++ semicolon.

• Simple data types: beginner-friendly names can be used, such as adad for integer and ashariya for floating-point values.

• Readable control flow: conditions and loops are expressed using Roman Urdu words while preserving clear block structure.

• LLVM-based code generation: valid programs are translated into LLVM IR, which can then be executed through JIT or compiled ahead of time.

• Compiler tools: Flex will perform lexical analysis and Bison will perform parsing and grammar validation.

4. Proposed Compiler Pipeline

Source Code	Flex	Bison	AST / Semantic Checks	LLVM IR	JIT / AOT
Roman Urdu program	Tokens	Grammar parsing	Validated program	Intermediate code	Execution / executable

5. Small Example

Example program written in the proposed Roman Urdu language:

```
adad x = 10 kardo
adad y = 20 kardo

agar (x < y) {
    dikhao("y bara hai") kardo
} warna {
    dikhao("x bara ya barabar hai") kardo
}

wapis 0 kardo
```

Roman Urdu	Meaning / Equivalent
adad	integer type
agar	if
warna	else
jabtak	while
dikhao	print/output
wapis	return
kardo	statement terminator (similar to ; in C/C++)

6. What Makes It Different

The project is not intended to copy C++ or another existing language word-for-word. Its distinguishing idea is a Roman Urdu-oriented syntax layer designed for readability and teaching. The compiler will define its own keywords and grammar, perform lexical and syntax analysis with Flex and Bison, and generate LLVM IR for execution. The initial implementation can support variables, arithmetic expressions, conditions, loops, printing, and functions, while leaving room for additional Roman Urdu constructs later.

7. Target Compilation Mode

The first target will be JIT compilation using LLVM so that Roman Urdu programs can be translated to LLVM IR and executed directly during testing. If time permits, the same LLVM IR can also be used for AOT compilation to produce a native executable.